THE DATA ENGINEER LAB

DATABRICKS FROM ZERO TO HERO

A 30-Day Interview & Career Readiness Series

DAY 5

Cluster Policies — Controlling Cost and Access at Scale

SECTION 01

What Is a Cluster Policy

A cluster policy is an admin-defined set of rules that controls what settings a user or group is allowed to choose when creating a cluster. Instead of giving every user unrestricted access to every configuration option, policies restrict the available choices to a safe, prescribed set.

CORE CONCEPT

A cluster policy is a tool used to limit a user or group's cluster creation permissions based on a set of policy rules. It specifies which attributes or attribute values can be used during cluster creation, and policies have ACLs that limit their use to specific users and groups.

LICENSING NOTE

Cluster policies require the Premium plan or higher. This is a common detail interviewers expect you to know when discussing governance features.

SECTION 02

Why Companies Use Cluster Policies

Cost Control

Policies can restrict which instance types are allowed and set a maximum DBU-per-hour limit for a cluster, directly preventing runaway compute costs from oversized or misconfigured clusters.

Simplified Self-Service

By fixing or hiding certain fields, policies let organizations give more users the ability to create their own clusters safely, without needing an admin to configure every cluster manually.

Governance and Consistency

Policies can enforce specific libraries, restrict which source is allowed to create a cluster (for example, only the Jobs service for job clusters), and limit the number of clusters a single user can have active at once.

SECTION 03

What a Policy Can and Cannot Control

Can Be Restricted	Cannot Be Restricted
Instance types and node counts	Library installation (handled by the Libraries API instead)
Max DBU-per-hour spend	Cluster permissions and ACLs (handled by a separate permissions API)

Number of active clusters per user	—
Cluster creation source (Jobs service vs Clusters UI vs REST API)	—

GOOD TO KNOW
Databricks recommends using compute policies rather than init scripts to install libraries in a controlled way, since policies can enforce cluster-scoped library installation as part of the same rule set.

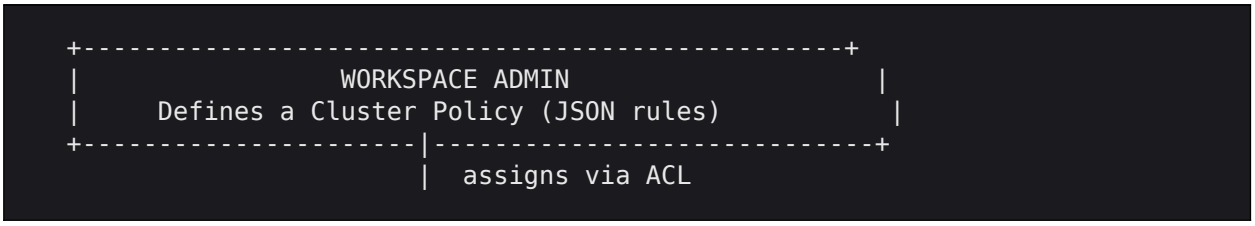
SECTION 04
What a Policy Definition Looks Like

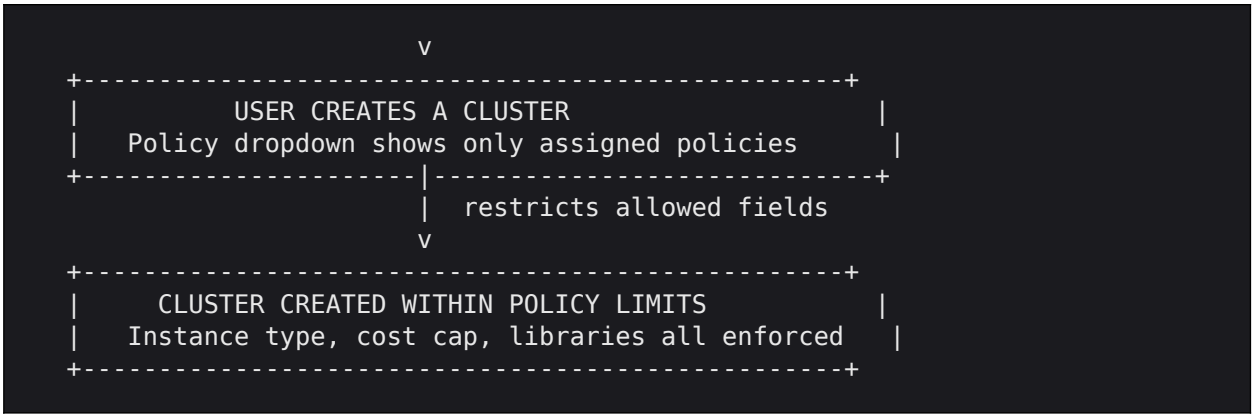
Cluster policies are written as JSON documents that define rules for specific cluster attributes. A simplified example restricting instance type and enforcing a cost ceiling:

```
{
  "node_type_id": {
    "type": "allowlist",
    "values": ["Standard_DS3_v2", "Standard_DS4_v2"]
  },
  "autotermination_minutes": {
    "type": "fixed",
    "value": 30
  },
  "dbus_per_hour": {
    "type": "range",
    "maxValue": 10
  }
}
```

This example allows only specific VM types, forces auto-termination after 30 minutes of inactivity, and caps the cluster's DBU-per-hour spend, all enforced automatically at cluster creation time.

SECTION 05
How Policies Fit Into Cluster Creation





REMEMBER THIS FOR INTERVIEWS

A cluster policy does not create a cluster; it constrains what a user is allowed to choose while creating one. The actual cluster creation is still done by the user, within the boundaries the policy sets.

SECTION 06

Policy Families

Policy families are Databricks-provided policy templates with pre-populated rules, designed to address common compute use cases. Instead of writing a policy definition from scratch, admins can start from a policy family and override only the parts they need to customize, using policy_family_definition_overrides.

- Personal Compute: a Databricks-managed default policy available on all workspaces, allowing users to create single-machine compute for individual use.
- Custom overrides: admins can inherit a policy family's rules and merge in their own additional restrictions.

SECTION 07

Glossary — Day 5 Terms

Term	Definition
Cluster Policy	An admin-defined set of rules limiting which cluster attributes and values a user can select at creation time.
Policy Definition	The JSON document that specifies the rules of a cluster policy.
Policy Family	A Databricks-provided policy template with pre-populated rules for common use cases.
DBU-per-Hour Limit	A synthetic policy attribute capping the maximum Databricks Units a cluster can consume hourly.
ACL (Access Control List)	The permission set that determines which users or groups can use a specific cluster policy.

Personal Compute	A Databricks-managed default policy enabling users to create single-machine compute for individual use.

SECTION 08

Interview Questions & Answers

Q1. What is a cluster policy in Databricks?

A. It is a tool that lets admins limit a user or group's cluster creation permissions using a set of rules, specifying which attributes or attribute values are allowed during cluster creation.

Q2. How do cluster policies help control cost?

A. Policies can restrict allowed instance types and set a maximum DBU-per-hour limit, which places a hard ceiling on how much a cluster created under that policy can cost per hour.

Q3. Can cluster policies control library installation?

A. Yes, policies can enforce cluster-scoped library installation as part of the rule set, and Databricks recommends this approach over using init scripts for installing libraries.

Q4. What is a policy family?

A. A Databricks-provided policy template with pre-populated rules for common compute use cases, which admins can inherit from and customize using definition overrides rather than writing a policy from scratch.

Q5. What can a cluster policy NOT control?

A. Library installation itself is handled by the Libraries API, and cluster permissions or ACLs are handled by a separate permissions API; these are not restricted directly through a policy definition.

Q6. What plan is required to use cluster policies?

A. Cluster policies require the Databricks Premium plan or higher.

STAY ON TRACK FOR ALL 30 DAYS

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